But why does rate–targeting work?

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Abstract

This is a short note on the target-rate objective introduced in [1]. We briefly discuss the mathematical heuristics of why this objective tends to discourage latent collapse in generative models, with vanilla VAEs as our main working paradigm.

Our main reference is [1]. There, the authors study posterior collapse through the lens of information theory, in particular through the *mutual information* $I_e(X;Z)$ between the input X and the latent variable Z. Loosely speaking, this quantity measures how much information about the input x is encoded in the latent variable z. More technically, it quantifies how far the encoder distribution $e(z \mid x)$ is from a distribution on z that is independent of x, namely the corresponding aggregated posterior; see also Section 1 & 2.

The central idea is that posterior collapse corresponds to the regime in which $I_e(X;Z)$ becomes small or vanishes, so that the latent variable carries little or no information about the input. By deriving variational bounds for the mutual information between the input X and the latent variable Z, the authors analyze when this quantity collapses to zero, what this means for the learned representation, and why the standard VAE objective might fail to prevent it. From this perspective, they also propose a practical modification of the loss, namely the *target-rate objective*.

Our discussion here focuses mainly on the practical implementation proposed in [1] and on the mathematical intuition behind why it helps. It is worth noting that the underlying viewpoint applies more broadly to latent-variable generative models, and is not restricted only to the standard VAE architecture.

1 Notation

$p^*(x)$	True data distribution.
$p_\theta(x)$	Model marginal over x, given by $$p_\theta(x) = \int p_\theta(z)\, p_\theta(x \mid z)\, dz,$$ and trained to approximate $p^*(x)$.
$p_\theta(z \mid x)$	Posterior induced by the generative model.
$q_\phi(z \mid x)$	Encoder, or approximate posterior.
$p_\theta(x \mid z)$	Decoder, or likelihood model.

$p_\theta(z)$ Prior. In standard VAEs, this is often fixed to $\mathcal{N}(0, I)$, while in [1] it may also be treated as learnable in the analysis.

$p_\theta(x, z)$ Generative joint distribution, given by

$$p_\theta(x, z) = p_\theta(z)\, p_\theta(x \mid z).$$

$q_\phi(z)$ Aggregated posterior, given by

$$q_\phi(z) = \int p^*(x)\, q_\phi(z \mid x)\, dx.$$

This is the latent-space distribution induced by the encoder and the data distribution.

2 Mutual Information and decompositions

Following the notation in [1], we choose an **encoder** $e(z|x) = q_\phi(z|x)$, **decoder** $d(x|z) = p_\theta(x|z)$ and **prior** $m(z) = p_\theta(z)$.

The joint distribution is given by $p_e(x, z) = p^*(x)e(z|x)$.

The aggregated posterior is $q_\phi(z) = p_e(z)$.

The posterior over the data under the encoder[1], $p_e(x|z) = \frac{p^*(x)e(z|x)}{p_e(z)}$.

Mutual information: $I_e(X; Z) := \iint p^*(x)\, e(z \mid x)\, \log \dfrac{p^*(x)\, e(z \mid x)}{p^*(x)\, p_e(z)}\, dz\, dx.$

Rate (KL-divergence): $R = \int p^*(x) \int e(z\,|\,x) \log \dfrac{e(z\,|\,x)}{m(z)}\, dz\, dx.$

Distortion (reconstruction loss): $D = -\int p^*(x) \int e(z\,|\,x) \log d(x\,|\,z)\, dz\, dx$

Entropy (treated as constant): $H = -\int p^*(x) \log p^*(x)\, dx.$

One can easily verify the following identities,

$$R \;=\; I_e \;+\; \underbrace{\mathrm{KL}\big(p_e(z) \,\|\, m(z)\big)}_{\delta \text{ (prior mismatch)}} \tag{1}$$

$$D \;=\; H - I_e \;+\; \underbrace{\mathbb{E}_{p_e(z)}\big[\mathrm{KL}\big(p_e(x \mid z) \,\|\, d(x \mid z)\big)\big]}_{\varepsilon \text{ (decoder mismatch)}} \tag{2}$$

Through those decompositions, one obtains the variational bounds regarding the mutual information

$$H - D \leqslant I_e(X; Z) \leqslant R.$$

These bounds allow us to view the optimization problem through the lens of rate–distortion theory.

[1] It gives the probability that actual data points x (not reconstructions, as it is for $d(x|z)$) produced that given z (under the chosen encoder model $e(z|x)$).

A particularly useful source of intuition is the feasible region in the rate–distortion plane. In the idealized infinite-capacity setting considered in [1], the variational bounds describe the boundary of this region and help explain both the shape of the RD curve and why ELBO alone is not sufficient to guarantee a meaningful latent representation.

A more detailed information-theoretic discussion, together with its connection to the standard β-VAE objective, can be found in [1].

In a more practical point of view, the training objective used in [1] is the *rate–target* loss

$$\mathcal{L}(e, m, d) \;=\; D(e, d) \;+\; \lambda\,|R(e, m) - R^*| \;=\; H - I_e + \epsilon + \lambda|I_e + \delta - R^*|. \tag{3}$$

This objective will ensure that the rate KL term is not going to zero.

Observe that in the typical case of VAEs

$$\mathcal{L}(e, m, d) = \mathcal{L}(e, d)$$

since the prior is a fixed distribution, e.g. $\mathcal{N}(0, I)$.

Question. Is this is enough to claim that we do not have posterior collapse (i.e. $e(z \mid x) \sim e(z)$[2])?

Not by itself. As we shall see, there are cases where we may have large KL divergence and at the same time posterior collapse.

3 How can a "fake" large KL be produced?

By "fake" we mean that R is large while $I_e \approx 0$ (the encoding carries little/no information about x), so the largeness must come from δ in (1). This only requires the aggregate posterior $p_e(z)$ to be far from $m(z)$. But if the goal is *fake* rate (keep $I_e = 0$), then making e independent of x is the direct route.

1. **Case 1: learnable prior.** If m is learnable, choose $m(z)$ far from $p_e(z)$; so $\delta = \mathrm{KL}(p_e(z) \,\|\, m(z))$ is large even with $e(z \mid x)$ independent of x (hence $I_e = 0$).
   For example, $e(z|x) = N(0, I)$ for all x. Then choose $m(z) = \mathcal{N}(\mu, I)$ for all x with $\|\mu\| \gg 0$.

2. **Case 2: fixed prior.** If m is *fixed*, choose $e(z \mid x)$ *independent of* x but with $p_e(z)$ far from m.
   For example, $m(z) = \mathcal{N}(0, I)$, then $e(z \mid x) = \mathcal{N}(\mu, I)$ for all x with $\|\mu\| \gg 0$ (Observe that under these assumptions, $e(z \mid x) = p_e(z)$).

In both cases, $I_e = 0$ and $R = \delta$ can be made arbitrarily large.

4 Why does this process not admit "fake" large KL (almost)?

With a learnable prior, prior mismatch δ can be reduced directly by adapting the marginal model (prior) m, which yields a cleaner mathematical argument. For a fixed prior, however, reducing δ requires changing the encoder itself, and this generally also affects the decoder inefficiency term ε. Therefore, in the fixed-prior setting, one obtains conditional statements.

[2] By $e(z)$ we mean a distribution that is independent of x; not to be confused with the aggregate posterior $p_e(z)$.

Case 1: learnable prior. From (1) and the rate-target loss (3), any $\delta > 0$ contributes to $|R - R^*|$ but does not appear in D. A gradient step on m that moves m toward p_e *always* reduces $\delta = \mathrm{KL}(p_e(z) \,\|\, m(z))$ and thus lowers the penalty term without changing D (only the δ term depends on the prior m). Therefore, with $\lambda > 0$, a solution relying on large δ cannot be a local optimum, i.e. optimization will shrink δ as much as the prior allows. After δ is small, the only way to reach $R \approx R^*$ is to increase I_e (make $e(z \mid x)$ depend on x); then reducing ε by letting $d(x \mid z)$ approximate $p_e(x \mid z)$ lowers D.

Case 2: fixed prior. For fixed prior $m = p(z)$, the target-rate objective can be written as

$$\mathcal{L}(e, d) = H - I_e(e) + \varepsilon(e, d) + \lambda |I_e(e) + \delta(e) - R^*|,$$

where

$$\delta(e) = \mathrm{KL}(p_e(z) \| p(z))$$

and

$$\varepsilon(e, d) = \mathbb{E}_{p_e(z)} \mathrm{KL}(p_e(x|z) \| d(x|z)).$$

A gradient step on the decoder d affects only ε, and hence only the distortion term. It does not directly change I_e, δ, or R. Therefore, decoder optimization alone does not rule out "fake" large KL.

A more useful viewpoint is to optimize the decoder for each fixed encoder and define

$$\varepsilon^*(e) := \inf_d \varepsilon(e, d).$$

This yields the reduced objective

$$\mathcal{L}^*(e) = H - I_e(e) + \varepsilon^*(e) + \lambda |I_e(e) + \delta(e) - R^*|.$$

If the decoder family is sufficiently expressive so that $\varepsilon^*(e)$ is small, then at rate $R \approx R^*$ the objective is approximately

$$H - I_e(e) + \lambda |I_e(e) + \delta(e) - R^*|.$$

In particular, among encoders satisfying $I_e + \delta = R^*$, the penalty vanishes and the objective is minimized by maximizing I_e, equivalently by minimizing δ. Thus, in this idealized fixed-prior setting, informative use of the latent variable is preferred over "fake" large KL.

References

[1] Alexander A. Alemi, Ben Poole, Ian Fischer, Joshua V. Dillon, Rif A. Saurous and Kevin Murphy: *Fixing a Broken ELBO*, Proceedings of the 35th International Conference on Machine Learning, **80**, pp. 159–168, 2018.